

QUEUE USING ARRAY

DATE:27-10-2025

CODE:

```
#include <stdio.h>

#define SIZE 5

int queue[SIZE], front = -1, rear = -1;

void enqueue(int value) {
    if (rear == SIZE - 1)
        printf("Queue is Full\n");
    else {
        if (front == -1) front = 0;
        queue[++rear] = value;
        printf("%d inserted\n", value);
    }
}

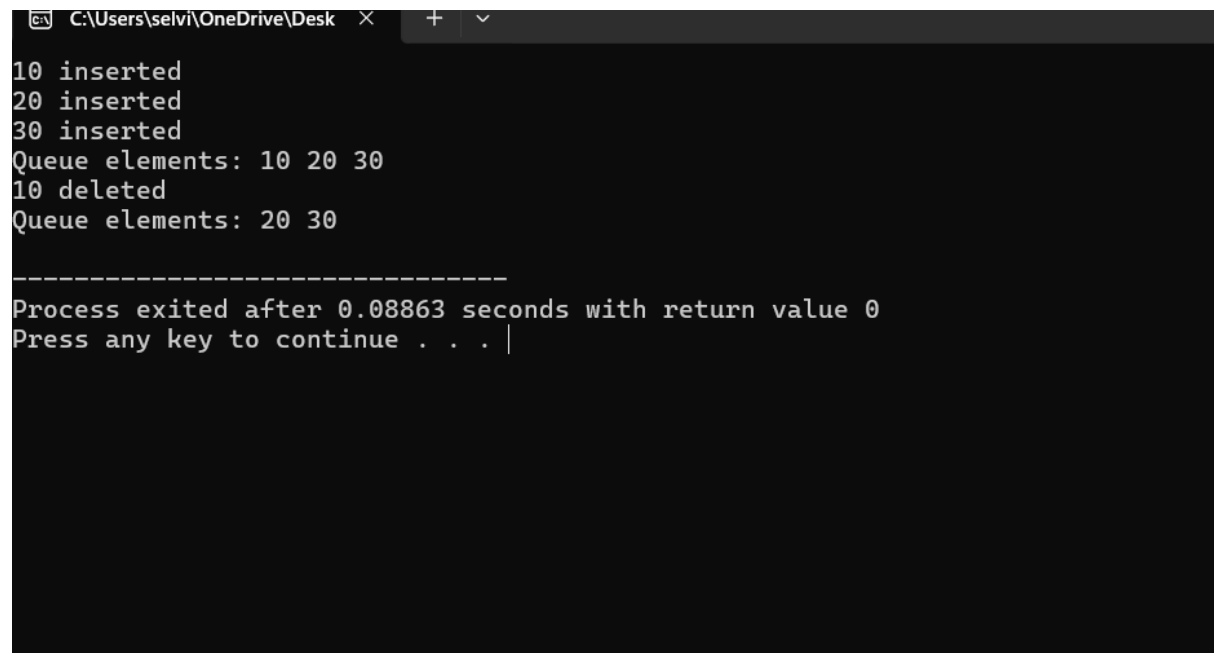
void dequeue() {
    if (front == -1 || front > rear)
        printf("Queue is Empty\n");
    else
        printf("%d deleted\n", queue[front++]);
}

void display() {
```

```
        int i=0;
    if (front == -1 || front > rear)
        printf("Queue is Empty\n");
    else {
        printf("Queue elements: ");
        for (i = front; i <= rear; i++)
            printf("%d ", queue[i]);
        printf("\n");
    }
}
```

```
int main() {
    enqueue(10);
    enqueue(20);
    enqueue(30);
    display();
    dequeue();
    display();
    return 0;
}
```

OUTPUT:



```
C:\Users\selvi\OneDrive\Desktop >
10 inserted
20 inserted
30 inserted
Queue elements: 10 20 30
10 deleted
Queue elements: 20 30

-----
Process exited after 0.08863 seconds with return value 0
Press any key to continue . . . |
```